Can a training intervention make a small LM show *increasing* returns to experience?

A sandbox discovery and its test on CHILDES

Standard Model 2 — exploratory side-arc

2026-06-20

1 The question

Children show *increasing* returns to linguistic experience: harder words are acquired *faster* later in development (the per-child scaling exponent $\kappa_i \approx 10$ on cumulative input). Language models, scaled over data, show the opposite — *diminishing* returns (our CHILDES developmental ladder gives an effective κ falling ∼1.3→0.5 over budget). Read as a *returns-to-experience* exponent, $\kappa < 1$ is diminishing (the LM default), $\kappa = 1$ a unit accumulator, and $\kappa > 1$ the kid-like "learning-to-learn" signature.

This side-arc asks a hypothesis-generation question: is there *any* training intervention that flips a small from-scratch LM from diminishing toward kid-like increasing returns — and if so, is it a genuine learning-to-learn effect or an optimization artifact? We answer in two phases: a fast sandbox sweep that *finds* a lever, and a controlled CHILDES experiment that *stress-tests* what it actually does.

Metric. For a run we probe per-word mean surprisal at ∼log-spaced steps and define

$$J = r_{\text{late}} - r_{\text{early}} = \big(\text{slope of } -\text{surprisal on } \log_{10}\text{experience, late half}\big) - \big(\text{same, early half}\big).$$

$J < 0$ = diminishing returns; $J \approx 0$ = unit accumulator; $J > 0$ = **increasing returns** (the acceleration signature). **Artifact controls:** $J > 0$ counts only if early learning stayed healthy ($r_{\text{early}} \gtrsim 0.4$) *and* final competence stayed near baseline — otherwise a model that simply learns slowly then "catches up" fakes $J > 0$ mechanically.

2 Phase 1 — Sandbox sweep (Karpathy `autoresearch`, ClimbMix GPT-2)

We repurposed a 5-minute single-GPU GPT-2 training loop (nanochat-style, ∼50M params, ClimbMix web text) to optimize for *acceleration* rather than loss, and swept **79 variants** across 12 intervention families (repetition, grokfast, weight decay, learning-rate magnitude/schedule, batch size, depth, init scale, optimizer, attention window, and combinations), scoring J with the artifact controls above.

Result: one robust lever — a flat, low learning rate. It is the only family that produced $J > 0$ while passing both artifact controls, as a clean monotonic dose–response (Fig. 1). The mechanism is a *conjunction*: increasing returns require a **flat (non-decaying) schedule** *and* a

low global LR magnitude — neither alone suffices (a decay schedule kills it at any magnitude; a flat schedule at full magnitude is also insufficient; verified with a flat-vs-decay $\times$ magnitude 2×2). Everything else was null or antagonistic: repetition strongly diminishes (overfitting); grokfast's $J > 0$ is a slow-start artifact (early learning collapses); schedule *shape*, depth, batch, and small-init alone do nothing.

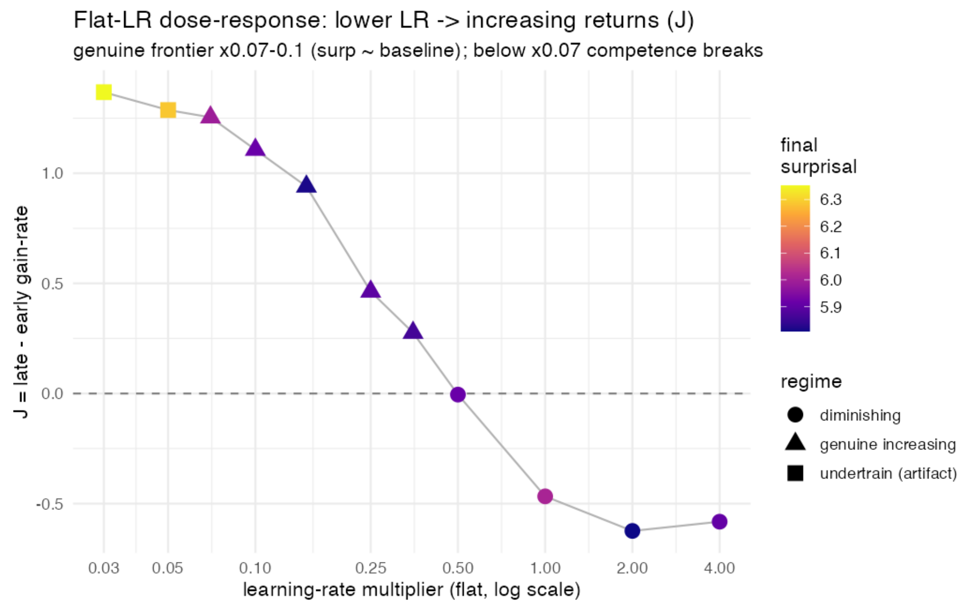

Figure 1: **Sandbox dose–response.** Lower flat LR $\rightarrow$ higher J at maintained competence (green = passes artifact controls). Below $\sim$0.07$\times$, J keeps rising but competence breaks (undertraining; orange). On this *web-text* run, the flat-low arms matched baseline on easy high-frequency words but cost $\sim$0.04–0.09 bits/token of overall validation loss — they trade endpoint optimization for a de-front-loaded trajectory.

The honest caveat from Phase 1. The flat low LR *mechanically* sustains late-training gains and de-front-loads the loss curve on the log-experience axis. By our J metric it is the genuine increasing-returns signature, but whether it is the *same mechanism* as children's increasing returns was left open — and it came with a validation-loss cost on web text. Phase 2 was designed to decide this on real child-directed data, and to add a structural-competence readout (grammar) the sandbox lacked.

3 Phase 2 — Test on CHILDES (GPT-2 small, 24M-word corpus)

We trained GPT-2 small from scratch on the full $\sim$24M-word CHILDES corpus (the Feng et al. setup, multi-epoch on child-directed speech) and compared a **standard decayed schedule** (linear decay from 10^{-4}) against a **flat constant schedule** at three magnitudes (10^{-4}, 3×10^{-5}, 10^{-5}), $\times$ 2 seeds, at two budgets: **1**$\times$ (20 epochs) and **4**$\times$ (60 epochs; the "never-finished" test). Along a shared log-spaced step axis we logged: validation `bits/token`; **CDI-word surprisal** $\rightarrow J$ (lexical-acquisition axis); **Zorro** (grammar, *primary* — its minimal pairs use a CHILDES-controlled vocabulary, so it isolates syntax from lexical coverage); and **BLiMP** (grammar, *secondary/comparability* only — vocabulary-mismatched to CHILDES, so its level is depressed and noisy). New tooling: a BLiMP/Zorro minimal-pair scorer and a grammar-trajectory callback, validated end-to-end.

Table 1: Endpoint values (mean over 2 seeds). Lower val_bpb = better; Zorro/BLiMP higher = better.

budget	schedule (LR)	val_bpb	Zorro	BLiMP	J
1×	decay 10^{-4} (standard)	2.064	0.694	0.576	+0.42
1×	flat 10^{-4}	2.225	0.674	0.576	+0.32
1×	flat 3×10^{-5}	2.030	0.701	0.577	+1.02
1×	**flat** 10^{-5}	**2.017**	0.690	0.589	+**1.60**
4×	decay 10^{-4} (standard)	2.673	0.689	0.576	−0.60
4×	flat 10^{-5}	2.106	0.689	0.580	+1.29

4 Results — four findings

(1) The flat-low-LR increasing-returns signature replicates on CHILDES. J rises monotonically as the flat LR drops: flat 10^{-5} ($J \approx 1.60$) > flat 3×10^{-5} (1.02) > flat $10^{-4} \approx$ decay (∼0.3–0.4) (Fig. 2). The sandbox dose–response holds on real data and a real model.

(2) On CHILDES, flat-low LR also *wins* validation loss — flipping the web-text cost. flat 10^{-5} reaches val_bpb 2.017 vs. the standard schedule's 2.064 (Fig. 3); flat *full*-magnitude is worst (2.225, overfits hardest). This is exactly the regime difference predicted: CHILDES is small-data *multi-epoch*, where a gentle low LR **overfits less**, so the sandbox's endpoint-loss *cost* becomes a *benefit*.

(3) The 4× test is decisive: the standard schedule catastrophically overfits; flat-low is robust. With more epochs, standard decay's val_bpb *worsens* 2.06 → 2.67 and its J flips +0.42 → −0.60 (it memorizes, then validation rises). Flat 10^{-5} barely moves (2.02 → 2.11, J stays +1.3) (Fig. 4). Flat-low does not "catch up" — it was already ahead — but it is the only schedule that survives extended training.

(4) Grammar is *decoupled* from all of this. Zorro plateaus at ∼0.68–0.70 for *every* arm, schedule, and budget (Fig. 5); BLiMP likewise sits at its CHILDES ceiling (∼0.58). The schedule dramatically reshapes the loss trajectory and the lexical returns (J), but **grammatical competence is insensitive to it** — it reaches its plateau early and stays.

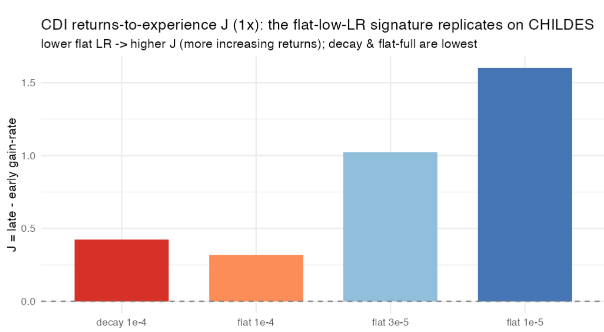

Figure 2: (1) J dose–response replicates.

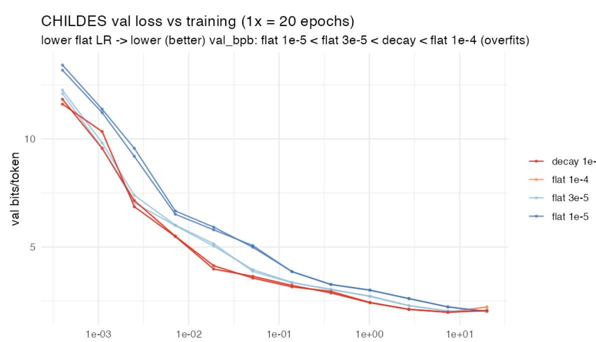

Figure 3: (2) flat-low wins val loss (anti-overfit).

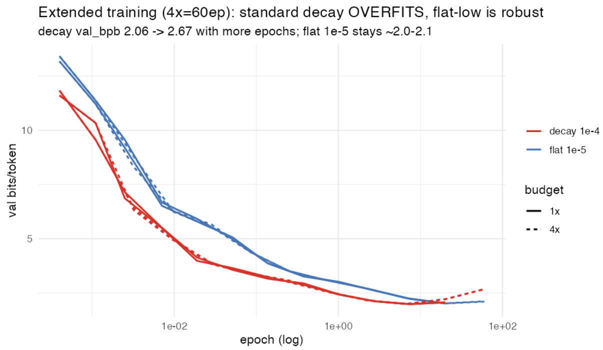

Figure 4: (3) 4×: decay overfits, flat-low robust.

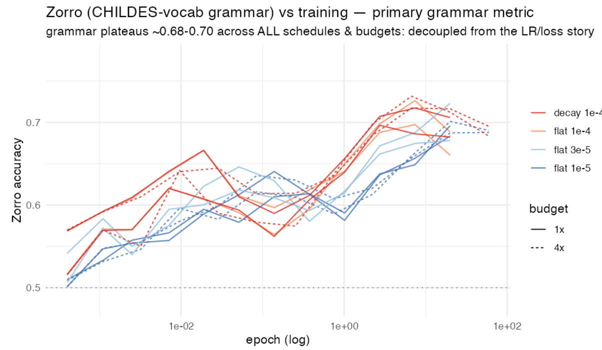

Figure 5: (4) grammar (Zorro) decoupled.

5 Interpretation

On CHILDES the flat-low-LR "acceleration" is, to first order, an **optimization / regularization** phenomenon: a gentle constant LR in the multi-epoch small-data regime *de-front-loads lexical accumulation and avoids overfitting*, which is why it simultaneously raises J, lowers validation loss, and is robust to extended training. But the decisive negative is finding (4): **none of this transfers to grammatical competence**. Zorro is flat across every arm. So the intervention does *not* make the model a better *structural* learner — it changes *when* and *how cleanly* the lexical distribution is absorbed, not what grammar is learned.

This sharpens the open question from Phase 1 in the useful direction. The kid-like reading of increasing returns is "learning-to-learn": earlier structure makes later structure easier, which should manifest as *compounding capability* (grammar, generalization). What flat-low LR delivers is increasing returns on the *lexical-accumulation* axis, **decoupled** from capability. For the paper's argument this is the clean statement: an LM *can* be pushed to show the kid-like J signature on word learning, but via an optimization knob that leaves grammatical competence untouched — evidence that the LM "acceleration" is mechanistically different from the child's, not just smaller.

6 Novel-word acquisition — a direct learning-to-learn test

The question left open above is whether the flat-low regime makes a *better learner* or merely a less-overfit model. We tested it directly, reusing the saved checkpoints. From each of the 12 checkpoints we warm-started and continued training under an **identical** protocol (same held-out CHILDES carrier, same constant LR 10^{-5}, same data order) on a corpus seeded with 24 **novel words** introduced by nonce-substitution (e.g. *blicket* placed in *car*'s contexts) at controlled frequency. The models have never seen these forms; we log per-nonce surprisal and take the **floor** reached after a fixed ∼100 exposures (lower = better learned). The decisive control: regress mastery on the source checkpoint's **J** (returns signature) *and* its starting **val_bpb** (competence) — if J predicts new-word mastery *beyond* competence, that is learning-to-learn, not just a healthier model.

Result (Fig. 6): new-word mastery tracks the regime, not competence. The floor correlates strongly with J ($r = -0.80$); in `lm(floor ~ J + val_bpb)` the J coefficient is significant (-2.51, $p = 0.026$) while val_bpb is not ($p = 0.51$). Flat-low (high-J) checkpoints learn genuinely new words to a lower floor than decay or flat-full, and the badly-overfit decay-4× checkpoints (high val_bpb but $J < 0$) are the *worst* new-word learners *despite the most training*. So the flat-low regime leaves a model that is specifically better at *acquiring new vocabulary*, dissociable from its overall loss. (A within-run variant — are later-introduced nonce cohorts learned faster? — is null once forgetting is controlled, as expected for a converged model re-seeing known data: the effect lives *across* regimes, not within one short continuation.) Caveat: 12 checkpoints, and J/val_bpb are correlated, so this is a strong-but-not-airtight dissociation; the simple r (-0.80 for J vs $+0.63$ for competence) is the most robust summary.

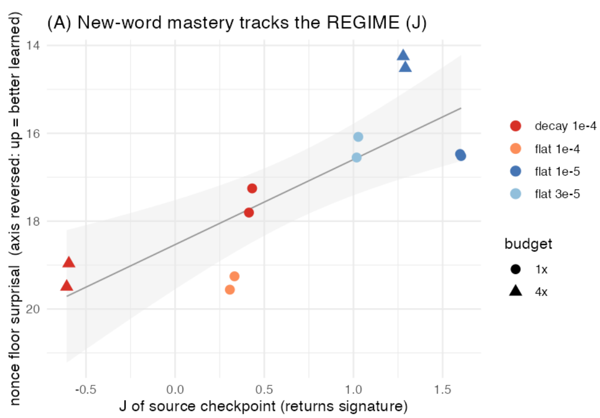

Figure 6: new-word mastery vs regime (J).

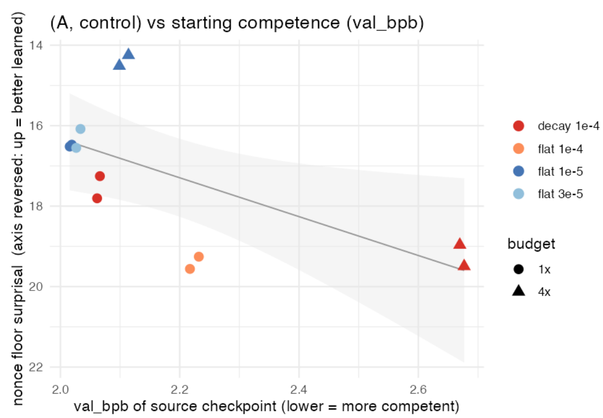

Figure 7: control: vs starting competence (weaker).

Refined verdict. Combining everything: the flat-low schedule **enhances the lexical-learning faculty specifically** — it yields the kid-like increasing-returns J on known words *and* makes the model a faster learner of genuinely new words (tracking the regime, not competence) — while leaving **grammatical competence untouched** (Zorro flat across all arms). "Lexical learning-to-learn, syntactically inert." That is a sharper and more interesting claim than either "it's just optimization" or "the LM accelerates like a child."

7 What this buys the paper / next steps

- A concrete, reproducible demonstration that the flat-low regime produces a genuinely better *word*-learner (lexical J *and* faster new-word acquisition, tracking regime not competence) but moves grammar not at all — a precise statement of how the LM "acceleration" is and isn't kid-like.
- The decisive learning-to-learn test is now *done* (§6) and positive; firm-ups: a second nonce-corpus seed (word-level CIs), a higher continuation LR to amplify, and a genuinely-novel-concept version (nonce words in *novel* roles, not fast-mapping into known slots).
- Worth a look: a CHILDES-vocabulary-*filtered* BLiMP for a sharper broad-syntax readout; and whether any intervention moves *grammar* trajectories at all (none here did).

Reproducibility

All code/data under `studies/llm/`: sandbox sweep + scorer in `autoresearch_pilot/` (`returns_score.R`, `bank_results.tsv`, `RESULTS.md`); CHILDES experiment in `childes_flatlr/` (`train_gpt2_childes_flatlr.py`, `eval_grammar.py`, `grammar_callback.py`, `run_arm.sh`, `dispatch_arms.sh`, `manifest_{1x,4x}.txt`, analyze.R, endpoints.csv). Runs on ccn2 at /data2/mcfrank/ladder/flatlr_grammar/runs/. 12 arms, 0 failures; Zorro = 23 CHILDES-vocab paradigms, BLiMP = 67 paradigms (minimal-pair log-prob scoring).